A Rank-Two Boundary Point Where γ_+ Is Not Locally Lipschitz

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Status. Candidate counterexample, likely valid, for human review.

1 Source question

Balan and Jiang define, for a Hermitian positive semidefinite matrix B,

$$\gamma_+(B) = \inf \left\{ \sum_j \|x_j\|_1^2 : B = \sum_j x_j x_j^* \right\}.$$

They prove continuity on the full positive semidefinite cone and recall local Lipschitz regularity on its positive definite interior. Remark 2.7 asks whether local Lipschitz regularity remains true on the entire cone [1, Remark 2.7, PDF p. 7].

Corollary 2.5. *Given $A \in \mathcal{S}_+^n \setminus \{0\}$, we have*

$$\gamma_+(A) \leq \sqrt{n} \cdot \min\left\{\mathrm{rk}(A), \sqrt{n}\right\} \cdot \|A\|_{1,1}.$$

Proof. The estimate follows directly from Propositions 2.3 and 2.4. □

Theorem 2.6. *γ_+ is continuous on $(\mathcal{S}_+^n, \|\cdot\|_{1,1})$.*

Remark 2.7. The continuity of γ_+ on the relative interior of $\mathcal{S}_+^n$ (the set of strictly positive definite matrices) was shown in [4]. However, this does not directly imply the continuity on $\mathcal{S}_+^n$. The paper [4] establishes additionally that γ_+ is locally Lipschitz on this relative interior. We do not know if this remains true on the entire $\mathcal{S}_+^n$.

Since $\mathcal{S}^n$ is a finite-dimensional vector space, it suffices to prove continuity with respect to $\|\cdot\|_{2\to2}$. The proof of Theorem 2.6 depends on the following lemma.

Lemma 2.8. *Let $A \in \mathcal{S}_+^n$ be of rank $r > 0$. Let $\lambda_r > 0$ denote the r-th eigenvalue of A.*

(A) Let $P_{A,r}$ denote the orthogonal projection onto the range or A. For any $\varepsilon \in (0,1)$ and $B \in \mathcal{S}_+^n$ with $\|A - B\|_{2\to2} \leq \frac{\varepsilon}{1-\varepsilon}\lambda_r$, we have

$$A - (1-\varepsilon)P_{A,r} B P_{A,r} \in \mathcal{S}_+^n.$$

(B) For any $\varepsilon \in (0, \frac{1}{2})$ and $B \in \mathcal{S}_+^n$ with $\|A - B\|_{2\to2} \leq \varepsilon\lambda_r$, we have

$$B - (1-\varepsilon)P_{B,r} A P_{B,r} \in \mathcal{S}_+^n,$$

where $P_{B,r}$ is the orthogonal project onto the top r eigenspace of B.

2 Claimed contribution

Theorem 1. *On the cone $\mathcal{S}_+^4(\mathbb{C})$, the functional γ_+ is not locally Lipschitz at*

$$A = \begin{pmatrix} 17 & -5 & -15 & 4 \\ -5 & 2 & 3 & -4 \\ -15 & 3 & 17 & 4 \\ 4 & -4 & 4 & 16 \end{pmatrix}.$$

The matrix A is positive semidefinite of rank two and $\gamma_+(A) = 125$. Consequently the question in Remark 2.7 has a negative answer.

The assertion is independent of the chosen norm on the ambient Hermitian matrix space, since that space is finite dimensional. In particular it holds for the entrywise ℓ_1 metric used in the source.

3 Proof intuition

At a locally Lipschitz point, a convex homogeneous functional has a finite supporting linear form. For γ_+, such a form is a Hermitian matrix T with $x^*Tx \le \|x\|_1^2$ for every vector x and $\operatorname{tr}(AT) = \gamma_+(A)$. We choose $A = VV^T$ so that the two-dimensional norm $z \mapsto \|Vz\|_1$ has three exact contact directions forming a weighted tight frame. Those contacts compute $\gamma_+(A)$ exactly. They also force T to calibrate three vectors. A linear relation among the vectors is incompatible with the three forced derivatives of the ℓ_1 norm, so the supporting form cannot exist.

4 Proof

We first isolate the convex-analytic implication used at the boundary.

Lemma 1 (Local Lipschitzness forces a supporting form)**.** *Let $f : \mathcal{S}_+^n(\mathbb{C}) \to [0, \infty)$ be convex and positively homogeneous. If f is locally Lipschitz at $A \in \mathcal{S}_+^n(\mathbb{C})$ relative to the cone, then there is a Hermitian matrix T such that*

$$f(B) \ge \operatorname{tr}(TB) \quad (B \in \mathcal{S}_+^n(\mathbb{C})), \qquad f(A) = \operatorname{tr}(TA).$$

For $f = \gamma_+$, this implies

$$x^*Tx \le \|x\|_1^2 \quad (x \in \mathbb{C}^n).$$

Proof. Choose a closed convex neighborhood $D = \mathcal{S}_+^n(\mathbb{C}) \cap \overline{B}(A, r)$ on which f is L-Lipschitz, and define on the real vector space of Hermitian matrices

$$g(X) = \inf_{Y \in D} \big(f(Y) + L\|X - Y\| \big).$$

This is the infimal convolution of the proper convex function $f + \mathbf{1}_D$ and the convex function $L\|\cdot\|$. Hence g is convex; it is finite on the whole ambient space because $A \in D$. The Lipschitz inequality shows that $g = f$ on D. Since a finite convex function on a finite-dimensional vector space has a subgradient at every point, there is a Hermitian T such that

$$g(X) \geq g(A) + \operatorname{tr}(T(X - A))$$

for every Hermitian X.

For $B \in \mathcal{S}_+^n(\mathbb{C})$ and sufficiently small $t > 0$, the point $A + t(B - A)$ belongs to D. The subgradient inequality and convexity of f give

$$\begin{aligned} f(A) + t\operatorname{tr}(T(B - A)) &\leq f(A + t(B - A)) \\ &\leq (1 - t)f(A) + tf(B). \end{aligned}$$

After division by t, this yields $f(B) \geq f(A) + \operatorname{tr}(T(B - A))$ for every B in the cone. Applying this to $B = 0$ and $B = 2A$, and using positive homogeneity, gives $\operatorname{tr}(TA) = f(A)$ and then $f(B) \geq \operatorname{tr}(TB)$.

Finally, $\gamma_+(xx^*) = \|x\|_1^2$: the one-term factorization gives the upper bound, while entrywise triangle inequality gives the reverse bound for every factorization. Substituting $B = xx^*$ proves the last assertion. $\square$

Now set

$$V = \begin{pmatrix} 4 & 1 \\ -1 & -1 \\ -4 & 1 \\ 0 & 4 \end{pmatrix}, \qquad H = \begin{pmatrix} 81 & 7 \\ 7 & 44 \end{pmatrix}.$$

Then $A = VV^T$ is the matrix in Theorem 1; in particular it is positive semidefinite of rank two.

Lemma 2 (Exact two-dimensional minorant)**.** *For every $z \in \mathbb{C}^2$,*

$$\|Vz\|_1^2 \geq z^* H z.$$

Equality holds for the three real directions

$$q_0 = (1, 0)^T, \qquad q_+ = (1, 4)^T, \qquad q_- = (-1, 4)^T.$$

Proof. First take a real vector (a, b). If $a = 0$, the inequality is immediate: $\|V(0, b)^T\|_1^2 = 49b^2 \geq 44b^2$. If $a \neq 0$, homogeneity and global sign reduce the assertion to $(a, b) = (1, t)$. Put

$$N(t) = \|V(1, t)^T\|_1 = |4 + t| + |1 + t| + |t - 4| + 4|t|, \quad Q(t) = 81 + 14t + 44t^2.$$

Directly on the five sign intervals,

t range	$N(t)$	$N(t)^2 - Q(t)$
$t \leq -4$	$-1 - 7t$	$5(t^2 - 16)$
$-4 \leq t \leq -1$	$7 - 5t$	$-19(t + 4)(t + 8/19)$
$-1 \leq t \leq 0$	$9 - 3t$	$-t(68 + 35t)$
$0 \leq t \leq 4$	$9 + 5t$	$19t(4 - t)$
$t \geq 4$	$1 + 7t$	$5(t^2 - 16)$

and every expression in the last column is nonnegative on its stated range. Equality occurs precisely at $t = -4, 0, 4$, which gives the three displayed directions.

For $z = u + iv$ with $u, v \in \mathbb{R}^2$, the Euclidean triangle inequality in $\mathbb{R}^2$ gives

$$\|Vz\|_1 = \sum_j \sqrt{(Vu)_j^2 + (Vv)_j^2} \geq \sqrt{\|Vu\|_1^2 + \|Vv\|_1^2}.$$

The real inequality applied to u and v therefore yields

$$\|Vz\|_1^2 \geq u^T Hu + v^T Hv = z^* Hz.$$

$\square$

The three contact directions satisfy the exact weighted frame identity

$$\frac{15}{16} q_0 q_0^T + \frac{1}{32} q_+ q_+^T + \frac{1}{32} q_- q_-^T = I_2. \tag{1}$$

Moreover,

$$\begin{aligned} x_0 &:= Vq_0 = (4, -1, -4, 0)^T, \quad \|x_0\|_1 = 9, \\ x_+ &:= Vq_+ = (8, -5, 0, 16)^T, \quad \|x_+\|_1 = 29, \\ x_- &:= Vq_- = (0, -3, 8, 16)^T, \quad \|x_-\|_1 = 27. \end{aligned}$$

Equation (1) provides a factorization of $A = VV^T$ whose cost is

$$\frac{15}{16} 9^2 + \frac{1}{32} 29^2 + \frac{1}{32} 27^2 = 125.$$

Thus $\gamma_+(A) \leq 125$.

Conversely, in any factorization $A = \sum_j y_j y_j^*$, every y_j belongs to $\operatorname{ran} A = \operatorname{ran} V$: this follows by testing the factorization against vectors in $\ker A$. Hence $y_j = Vz_j$ for some $z_j \in \mathbb{C}^2$, and the injectivity of V gives $\sum_j z_j z_j^* = I_2$. Lemma 2 now gives

$$\sum_j \|y_j\|_1^2 \geq \sum_j z_j^* H z_j = \operatorname{tr}\left(H \sum_j z_j z_j^* \right) = \operatorname{tr} H = 125.$$

Therefore

$$\gamma_+(A) = 125. \tag{2}$$

Suppose, toward a contradiction, that γ_+ is locally Lipschitz at A. Lemma 1 supplies a Hermitian matrix T with

$$x^* T x \leq \|x\|_1^2 \quad (x \in \mathbb{C}^4), \qquad \operatorname{tr}(AT) = 125.$$

Let $K = V^* T V$. Applying the inequality to Vq and using (1),

$$\begin{aligned} 125 = \operatorname{tr} K &= \frac{15}{16} q_0^* K q_0 + \frac{1}{32} q_+^* K q_+ + \frac{1}{32} q_-^* K q_- \\ &\leq \frac{15}{16} \|x_0\|_1^2 + \frac{1}{32} \|x_+\|_1^2 + \frac{1}{32} \|x_-\|_1^2 = 125. \end{aligned}$$

All three weights are positive, so equality holds in all three vector inequalities:

$$x_\bullet^* T x_\bullet = \|x_\bullet\|_1^2 \quad (\bullet \in \{0, +, -\}).$$

At such an equality vector, the nonnegative function $x \mapsto \|x\|_1^2 - x^*Tx$ has a minimum. Real and imaginary perturbations in any nonzero coordinate show

$$(Tx)_j = \|x\|_1 \operatorname{sign}(x_j) \quad \text{whenever } x_j \neq 0.$$

The second coordinate is negative in each of x_0, x_+, x_-, so

$$(Tx_0)_2 = -9, \qquad (Tx_+)_2 = -29, \qquad (Tx_-)_2 = -27.$$

But $x_+ - x_- = 2x_0$. Applying T and taking the second coordinate would give

$$-29 - (-27) = 2(-9),$$

that is, $-2 = -18$, a contradiction. This proves Theorem 1.

5 Verification and limitations

The exact symbolic checker in `code/verify_counterexample.py` verifies $A = VV^T$, $\operatorname{rank} A = 2$, all five polynomial factorizations in Lemma 2, the tight-frame identity, the factorization cost 125, and the final contact-vector contradiction. It runs with:

```
conda run --no-capture-output -n sandbox python \
  code/verify_counterexample.py
```

The proof gives a negative answer in dimension four. It does not classify which boundary points retain local Lipschitz regularity and does not decide whether a dimension-three counterexample exists.

6 Novelty check

The bounded search on 19 July 2026 covered the run's registry, solution, attempt, and proof-gap indexes; exact searches for arXiv:2409.20372, its title, the phrase "locally Lipschitz on the entire positive semidefinite cone," and the core terms `gamma_+`, rank-one decomposition, and Lipschitz boundary regularity. We checked the source paper, its 2021 predecessor [2], and a January 2026 author presentation that still labels the Lipschitz problem open [3]. No later solution or counterexample was found. This supports, but cannot certify, novelty.

References

[1] R. Balan and F. Jiang, *Factorization of positive-semidefinite operators with absolutely summable entries*, arXiv:2409.20372 (2024), `https://arxiv.org/abs/2409.20372`.

[2] R. Balan, K. A. Okoudjou, M. Rawson, Y. Wang, and R. Zhang, *Optimal ℓ^1 rank one matrix decomposition*, in *Harmonic Analysis and Applications*, Springer (2021), 21–41.

[3] R. Balan, *Factorization of positive-semidefinite operators with absolutely summable entries*, ILAS/JMM presentation, January 2026, `https://math.umd.edu/~rvbalan/PRESENTATIONS/RaduBalan_JMM2026_ILAS.pdf`.